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Effect of vegetation patchiness on the subsurface water distribution in abandoned farmland of the Loess Plateau, China

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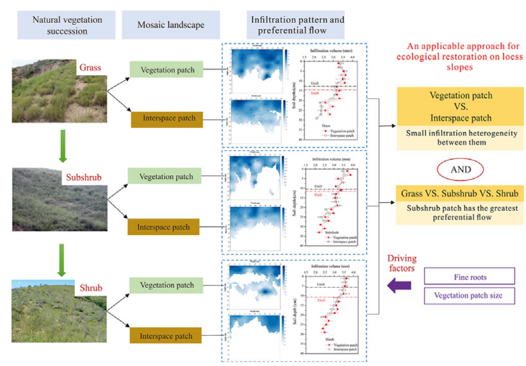
HIGHLIGHTS

- Vegetation patchiness affects soil properties and root systems.
- Vegetation patch increases preferential infiltration but not significantly.
- Subshurb exhibits the higher preferential flow infiltration than grass and shrub.
- Subsurface water flow is controlled by fine roots and vegetation patch size.

GRAPHICAL ABSTRACT



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ABSTRACT

Patchiness of grassland results in important effects on ecohydrological processes in arid and semiarid areas; however, the influences on subsurface water flow and soil water distribution remain poorly understood, particularly during vegetation succession on slopes. This study examined these effects by comparing the water flow behaviors and preferential infiltration between vegetation patches (VP) and interspace patches (IP) in three sites at different states of vegetation succession (grass, subshrub and shrub) in abandoned farmland of the Loess Plateau, China. Dye tracer infiltration showed that patchiness of vegetation increased spatial variations of soil water and preferential infiltration by increasing the densities of fine root length and fine root volume in the soil profile. Moreover, the more abundant and intricate roots following a lateral direction beneath VP likely contributed to lateral flow and infiltration variability. However, differences between VP and IP were not significant because considerable living fine roots and decayed roots of IP also provided preferential flow pathways. Our finding indicated that IP could compete with VP for access to soil water resources, which potentially increased hillslope-scale infiltration and reduced surface runoff and erosion risk. Under the different states of vegetation succession, subshrub patches showed significantly greater preferential infiltration volume (28.53 mm) and contribution of preferential infiltration to total infiltration (60.58%) than grass and shrub patches. Vegetation patch size made positive effects on improving preferential flow and water movement. Greater preferential flow in subshrub patches played a positive role in soil water storage and replenishment. Therefore, natural restoration of a slope area with small heterogeneity in preferential flow can be successfully applied in the Loess Plateau, particularly during the subshrub succession state.

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1. Introduction

A total of approximately 41% of the Earth's surface is comprised of arid and semiarid regions (Reynolds et al., 2007). These regions support water-limited ecosystems characterized by a patchy landscape of vegetation interspaced with open areas of relatively bare soil (Aguiar and Sala, 1999; D'Odorico et al., 2012; Wu et al., 2016). Such mosaic landscape is a source-sink system for the redistribution of soil resources, in which the spatial unit of interspace patch (IP) acts to generate runoff and sediment whereas the vegetation patch (VP) acts as a sink (Ludwig et al., 2005; Puigdefábregas, 2005; Merino-Martin et al., 2012). Previous studies have linked patchy vegetation to soil erosion, high evapotranspiration, reduced ecosystem function and land degradation (Schlesinger et al., 1999; Scott et al., 2006), whereas recent studies found that shrub encroachment appeared to enhance ecosystem function by increasing soil microbial functions and altering soil physicochemical properties (Eldridge and Soliveres, 2015; Chandregowda et al., 2018). Water availability is the key driver of productivity in arid and semi-arid environments; therefore, the role of vegetation patchiness in regulating the hydrological cycle and ecosystem balance has received increasing attention (Rossi and Ares, 2017; Barbosa-Briones et al., 2019; Liu et al., 2019).

Infiltration of water through soil is a fundamental process that influences the water budget of plant communities as well as the amount of surface runoff and consequently the risk of soil erosion by water (Chartier et al., 2011; Lu et al., 2019). A key condition for the formation and characteristic of vegetation pattern tended to be the infiltration contrast between VP and IP (Saco et al., 2007; Meron, 2018). In turn vegetation patterns also directly affect infiltration amount and spatial variability (Barbosa-Briones et al., 2019). The majority of previous studies have concluded that VP characterized by litter and herbage cover generally demonstrate elevated rates of infiltration due to decreased soil bulk density and increased soil nutrients and porosity (Puigdefábregas, 2005; Ravi et al., 2008; Mayor-Ángeles et al., 2009). The IP that was often covered by soil crusts induced decrease in water infiltration (Eldridge et al., 2000; Meron, 2016). Although VP and IP exhibit differences in soil surface condition and infiltration capacity (Ludwig et al., 2005; Peng et al., 2013), the differences in water transport and distribution through soil between the two patches remain poorly understood. The behavior of water flow is increasingly regarded as the primary driver of soil water distribution (Jiang et al., 2015). Soil water flow can be characterized into matrix flow and preferential flow (Jarvis, 2007), with the former referring to gradual and regular movement of solutes and water through the soil (Alaoui and Goetz, 2008), whereas the latter is rapid soil water flow through soil macrospores that bypass the majority of the soil matrix (Allaire et al., 2009). Preferential flow pathways allow water and solutes to move to greater soil depths than that estimated by the Richard's Equation (Beven and Germann, 1982; Benegas et al., 2014; Zhang et al., 2017), and are critical for facilitating replenishment of water in the root zone. Despite constituting a limited proportion of total soil volume, preferential flow may contribute between 11%–94% to total infiltration (Stumpp and Maloszewski, 2010; Alaoui, 2015; Liu et al., 2018; Zhang et al., 2018).

The canopy of patchy vegetation effectively acts as a funnel to channel stemflow to the root area (Li et al., 2009). On the other hand, the well-developed root system under VP generally increased macroporosity and soil aggregation, thus facilitating preferential flow pathways (Ghestem et al., 2011). Nevertheless, some studies have reported clogging of the soil pores and subsequent blocked water flow within a high-density root network (Archer et al., 2002; Cui et al., 2019). The effects of plant roots on preferential flow pathways are highly dependent on various root traits, including length, volume, orientation, decay rate (Jiang et al., 2018), distribution in soil profiles (Nespoulous et al., 2019) and complex root-soil interactions (Zhang et al., 2018). Therefore, understanding the differences in soil water flow behavior between VP and IP requires the monitoring and quantification of the relationship between plant roots and preferential flow paths in response to vegetation patchiness. Furthermore, the effect of vegetation patchiness on infiltration ultimately depends on the functional traits of plants, their root systems and the soil structure (Yu et al., 2017; Luo et al., 2019). Previous studies focused on the effects of different land cover or management practices on subsurface water flow (Jiang et al., 2015; Mei et al., 2018), whereas only a few studies have attempted to investigate the soil water infiltration under vegetated patches with different patch characteristics, such as patch size, shape and orientation (Rossi et al., 2018).

Most of the previous studies on the function of vegetation patchiness have been performed in flat terrain or gentle slopes, whereas relatively few studies have been undertaken on steeper slopes (Barbosa-Briones et al., 2019). The Chinese Loess Plateau has experienced considerable improvements in vegetation cover since 1999 when the "Grain to Green" project was initiated (Yu et al., 2019). An important component of the project was the intentional abandonment of vast areas of farmland on steeper slopes (Wang et al., 2019). It is probable that local plant communities will re-colonize abandoned farmland after natural restoration (Kou et al., 2016). Previous studies showed that soil physicochemical properties (i.e. soil organic matter, water stable aggregates) and vegetation properties (i.e. root systems) increased as vegetation succession progressed on abandoned farmland (Wang et al., 2019; Zhang et al., 2019b). The scarcity of water generally results in the distribution of vegetation following a heterogeneous pattern with a highly mosaic structure of patchy vegetation (VP) and patches of relatively bare ground (IP) (Hou et al., 2014). This mosaic pattern has significant effects on ecohydrological processes and on the risk of soil erosion in the sloping grassland of the Loess Plateau (Hao et al., 2016).

The present investigation conducted a dye tracer experiment to study the vertical soil water flow and preferential infiltration under VP and IP. Three abandoned farmland sites in the Loess Plateau restored to different states of vegetation succession were studied. The states of vegetation succession were grass, subshrub and shrub, under which the main plant communities were *Bothriochloa ischaemum*, *Artemisia sacrorum* and *Periploca sepium Bunge*, respectively. The objectives of the present study were to: (1) compare the soil water flow behaviors and preferential flow between VP and IP within vegetation states; (2) compare the water infiltration patterns among the three vegetation states and; (3) identify factors controlling differences in preferential flow and infiltration volume. Based on the current study, the following hypotheses are proposed: (1) a greater extent of water flow and preferential infiltration is expected in VP compared to IP since VP is characterized by a well-developed root system, and; (2) the degree of preferential flow will improve during the vegetation succession with increasing soil properties and root systems.

2. Materials and methods

2.1. Study area

The present study was conducted in the Fangta small catchment (36°46′13″–36°49′18″ N, 109°15′22″–109°17′01″ E) in An'sai County, located on the Loess Plateau, North China (Fig. 1). The area falls within a semi-arid monsoon climate zone. The annual temperature ranges from −23.6 °C to 36.8 °C with an average value of 8.8 °C (Wang et al., 2017). The mean annual rainfall is 505 mm, and more than 70% of the rainfall occurs from July to September (Wang et al., 2019). The soils in the experiment site are categorized as Calcaric Regosols, with mean clay, silt and sand contents of 11.32%, 24.31% and 64.87%, respectively. The catchment is located in a forest and grass ecotone. The implementation of the "Grain to Green" project has resulted in most cropland on slopes being abandoned for gradual artificial and natural restoration. The dominant natural plant species include *Stipa bungeana*, *A. gmelinii*, *Bothriochloa ischaemum*, *A. giraldii*, *Lespedeza davurica* and *Rosa xanthine*. The artificial vegetation types are *Hippophae rhamnoides*,

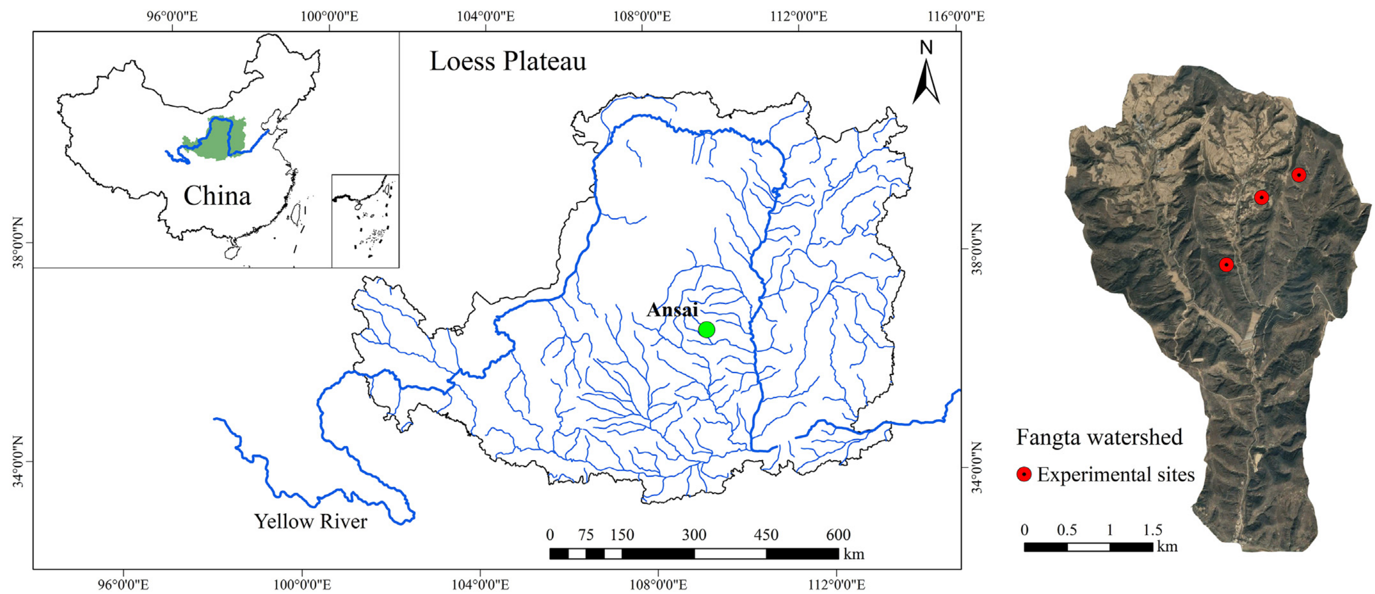

Fig. 1. Location of the study area.

Caragana korshinski and *Robinia pseudoacacia*. Biological soil crusts that mainly include cyanobacteria crust and moss crust distributed on the abandoned farmland in this watershed (Wang et al., 2017).

2.2. Experimental design

Three sites that have undergone vegetation restoration were chosen according to the state of vegetation succession (Figs. 1) based on field surveys of abandoned farmland in the study area: (1) site G containing a perennial herbaceous community (*Bothriochloa ischaemum*) that has been abandoned for 18 years; (2) site S1 containing a subshrub community (*Artemisia sacrorum*) that has been abandoned for 12 years and; (3) site S2 containing a shrub community (*Periploca sepium Bunge*) that has been abandoned for 25 years. Three plots with dimensions of 5 m × 5 m established on the middle slopes at each site were used to investigate the terrain, vegetation cover, distribution of VP and IP and number of patches (Table 1). In each site, three VP and IP, respectively of similar size and condition were chosen as replicated subplots. The selected subplots were separated by at least 2 m to prevent interference. Field surveys and sampling were conducted from July to August 2018, and dye tracer experiments were performed in subplots in which there had been no rainfall for five days.

2.3. Investigations of vegetation patch and interspace patch

VP and IP in the three plots were identified according to the definition of a vegetation patch by Muñoz-Robles et al. (2011). For each patch, attributes including area (m^2), length (m), width (m), height (m), perimeter (*Per*, m), compactness (*Com*) and geometric aspect ratio (*AR*) were estimated (Rossi et al., 2018). Aboveground biomass (AGB) for each subplot was estimated by sampling live vegetation rooted to the ground surface. The vegetation patch compactness (*Com*) corresponding to patch arrangement related to space was calculated as:

$$Com = \frac{Per}{2 \times \sqrt{\pi \times area}} \tag{1}$$

where *Per* and area are the vegetation patch perimeter (m) and patch area (m^2), respectively.

The *AR* of a vegetation patch was used as an indicator of patch elongation and was calculated as the ratio of patch length to patch width:

$$AR = \frac{Length}{width} \tag{2}$$

where *length* (m) and *width* (m) are the longer and shorter sides of the vegetation patches, respectively. The patch length was measured along the slope, and the width was measured along the contour line.

2.4. Dye tracer infiltration

A total of 18 dye tracer tests were conducted at the three sites (three sites × two patches × three subplot replicates). Each subplot was prepared by clipping aboveground plants to ground level and carefully removing all litter. A hollow stainless-steel cube (length × width × height of 0.4 m × 0.4 m × 0.4 m) was inserted 0.10 m–0.15 m into the soil (Fig. 2a). Each cube was irrigated with tap water (10 L) dyed with Brilliant Blue FCF (4 g L^{-1}). At completion of irrigation, a plastic sheet was placed over the cube to prevent evaporation and disturbance. After 24 h to allow for infiltration, the sheeting and stainless-steel cube were carefully removed. A vertical soil profile (width × height of 0.6 m × 0.6 m) was extracted at the center of the cube area across the slope, following which the soil profile was photographed using a Nikon digital camera (AF-S D5600, Nikon, Tokyo, Japan) under conditions of daylight (Fig. 2b).

The software packages ERDAS IMAGINE v9.2 and Matlab 2015a were used to process the images of soil profiles to quantify the dyed area as per procedures by Cey and Rudolph (2009). First, the images were geometrically corrected to classify the soil profile into non-dyed and dyed areas. The images were then set to a resolution of 400 × 400 pixels and saved to a bitmap format (Fig. 2c). Preferential flow indices were calculated based on the processed images to determine the soil flow patterns as follows (Fig. 2d):

Dye coverage (DC, %) defined as the percentage of the dyed area to the total area was expressed as follows:

$$DC = 100 \times \left(\frac{D}{D + ND}\right) \tag{3}$$

where D and ND are the areas of the dyed and non-dyed regions (cm^2), respectively.

Preferential Flow Proportion (PFP, %) represents the percentage of total infiltration flowing through paths of preferential flow, and is calculated by:

Table 1
Summary of experimental site information.

Site	Altitude (m)	Slope Aspect (°)	Slope Gradient (°)	Vegetation Coverage (%)	Patch density (/m²)	Dominat plant species in vegetation patches	Dominat plant species in interspace patches
G	1254	NE65	20	37	0.40	*Bothriochloa ischaemum*	*Artemisia sacrorum, Artemisia giraldii*
S1	1233	NE55	18	44	0.32	*Artemisia gmelinii*	*Artemisia argyi, Potentilla acaulis, Dendranthema indicum*
S2	1287	NE30	16	47	0.24	*Periploca sepium Bunge*	*Artemisia gmelinii, Stipa bungeana, Astragalus scaberrimus*

Note: Patch density refers to number of vegetation patches per plot with 25 m^2.

$$PFP = 100 \times \left(1 - \frac{Unifr \cdot 40}{TDA}\right) \tag{4}$$

where Unifr is the uniform infiltration depth (cm), defined as the depth at which there is a decrease in dye coverage to <80%, and was multiplied by the 40 cm wide profile in the present study and TDA stands for the total dyed area (cm^2).

The length index (LI) represents the sum of the absolute differences between the consequent DC values with depth in the soil profile as follows:

$$LI = \sum_{i=1}^{n} |DC_{i+1} - DC_i| \tag{5}$$

where DC_i is the dye coverage at the i-th soil layer; n is the number of pixels in the vertical profile, which is 400 based on the image resolution.

2.5. Soil sampling and analysis

Soil samples were collected next to the dye subplot before the measurements of infiltration. Soil sampling involved three replicates from a soil depth of 0 cm–40 cm at 10 cm intervals. Measured soil properties

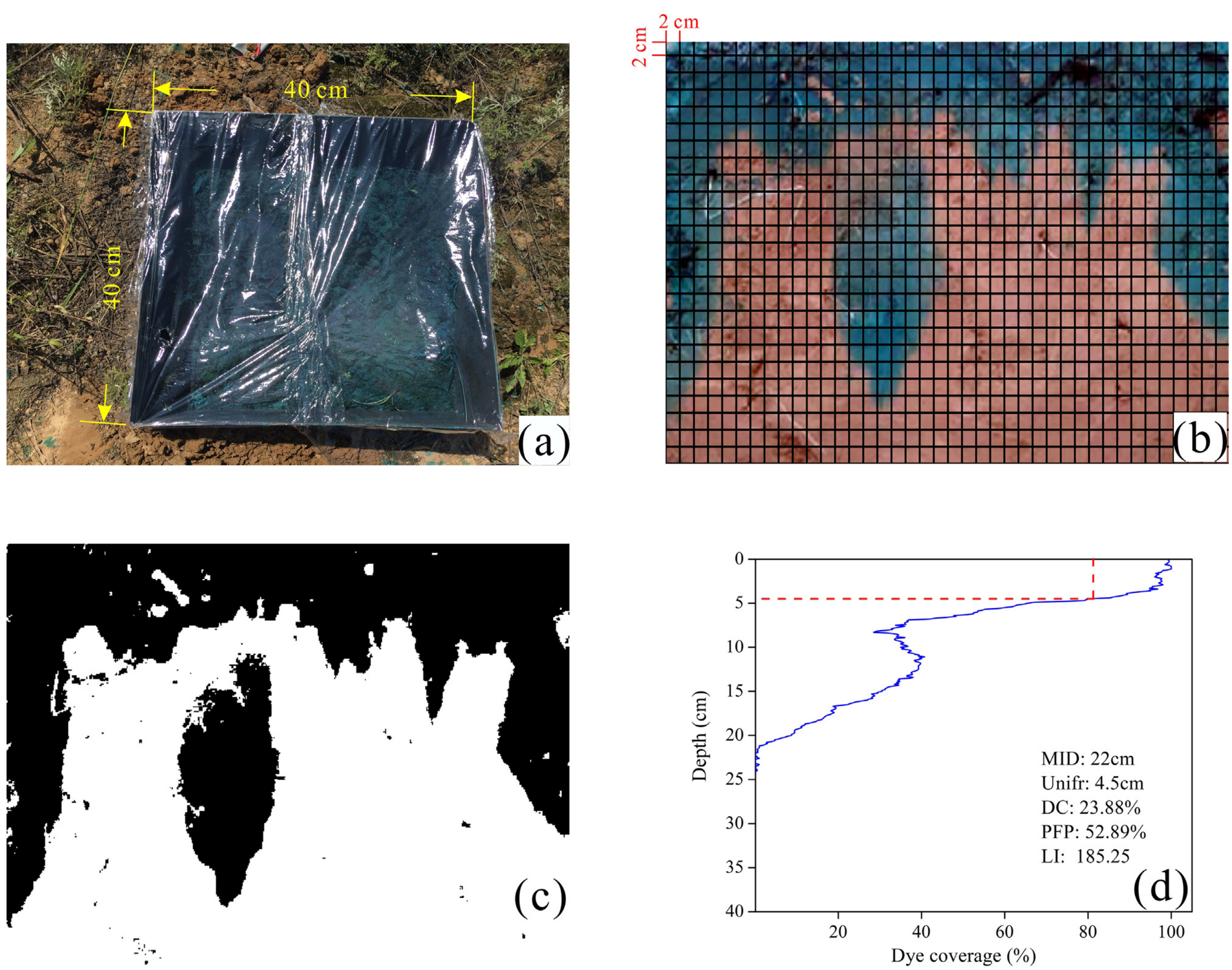

Fig. 2. Example of field dye tracer experiment by using the hollow stainless-steel cube (a), the raw image for dye stained profile and sampling (b), binary digital image (c), and corresponding parameters extraction (d).

included initial soil water content (ISWC, %), soil bulk density (BD, g cm^{-3}), soil particle size and the soil organic matter (SOM, g kg^{-1}). ISWC and BD were measured using the oven-drying method (105 °C for 24 h). Soil particle size was measured using a Mastersizer 2000 laser diffraction device (Malvern, United Kingdom), and soil particles were categorized by size as clay (<0.002 mm), silt (0.002 mm–0.05 mm) or sand (0.05 mm–2 mm). The soil aggregate distribution was measured using the wet sieve method and the weight proportion method was used to calculate the water-stable aggregate content >0.25 mm ($WR_{0.25}$). The potassium dichromate colorimetric method was used to measure SOM.

The spatial redistribution of soil water after irrigation was analyzed by dividing the soil profile into small grids (width × height of 2 cm × 2 cm) from which soil samples of 4 cm^3 (width × height × thickness of 2 cm × 2 cm × 1 cm) were collected (Fig. 2b). Soil samples totaling 400 were collected for each profile (width × height of 40 cm × 40 cm). After analysis for dye staining, the samples were dried at 105 °C for 24 h for measurement of soil gravimetric water content (SWC, %). Total infiltration volume (TIV, mm), preferential infiltration volume (PIV, mm) and the contribution of PIV to TIV (Con, %) was then calculated as follows:

$$\mathrm{TIV} = \sum_{i=0}^{i=\mathrm{MID}} \mathrm{Q_i} \tag{6}$$

$$\mathrm{PIV} = \mathrm{TIV} - \sum_{j=0}^{j=\mathrm{Unifr}} \mathrm{Q_j} \tag{7}$$

$$\mathrm{Q} = 10\sum_{k=1}^{k=20} (\mathrm{SWC_{ak}} - \mathrm{SWC_{bk}})/40 \tag{8}$$

$$\mathrm{SWC_k} = \mu_\mathrm{k} \times \mathrm{BD_k} \times \mathrm{v} \tag{9}$$

$$\mathrm{Con} = 100 \times \left(\frac{\mathrm{PIV}}{\mathrm{TIV}}\right) \tag{10}$$

where TIV (mm) is the total infiltration volume, MID (cm) is the maximum infiltration depth, Q_i and Q_j (mm) are the infiltration volumes at soil layers *i* and *j*, respectively, PIV (mm) is the preferential infiltration volume, Unifr (cm) is the uniform infiltration depth, SWC_{bk} and SWC_{ak} (mm) represent the soil water content before and after irrigation, respectively, *k* is the number of the sequence of the horizontal soil sample, and *k* has a maximum value of 20 as soil was sampled every 2 cm within each 40 cm wide horizontal soil profile, μ_k (%) is the *k*-th sample's gravimetric soil water content, BD_k (g cm^{-3}) is the soil bulk density of the layer, *v* (cm^3) is the volume of the soil sampled (cm^3) and was 4 cm^3 and Con (%) is the proportion of PIV to TIV.

2.6. Root measurements

After the dye tracer experiment, a rectangular metal box (length × width × height of 20 cm × 10 cm × 10 cm) was used to delineate the area for sampling intact soil containing roots in each subplot. Root samples with three replicates were collected for four soil layers in each soil profile: (1) 0 cm–10 cm; (2) 10 cm–20 cm; (3) 20 cm–30 cm and; (4) 30 cm–40 cm. The samples were placed in a sealed plastic bag and stored under refrigeration for 12 h at 4 °C. During analysis, the intact soil samples with roots were washed with tap water and the roots were carefully extracted. Root parameters, including root diameter (cm), root length (cm), root surface area (cm^2) and root volume (cm^3), were measured by using a V700 scanner (Epson, Tokyo, Japan) and Regent WinRhizo software (2009a, Regent Instruments, Quebec, QC, Canada). Roots were categorized into the categories of fine and course roots with diameter ≤ 5 mm and diameter > 5 mm, respectively. All roots were oven-dried for 24 h at 65 °C, following which they were weighed to obtain an estimate of belowground biomass. Root mass density (RMD, mg cm^{-3}), coarse root length density (RLD_{coarse}, cm cm^{-3}), fine root length density (RLD_{fine}, cm cm^{-3}), coarse root volume density (RVD_{coarse}, cm^3 cm^{-3}) and fine root volume density (RVD_{fine}, cm^3 cm^{-3}) were calculated according to Zhou and Shangguan (2007).

2.7. Statistical and spatial analyses

Before statistical analysis, normality tests for all data were conducted using the Shapiro-Wilk method and all data were normally distributed. Two-way analysis of variance (ANOVA) was carried out to assess the differences of preferential flow indices, soil characteristics, root traits and vegetated patch attributes between the two patches and among the three sites. All of the significant differences were determined at the level of $p < 0.05$ using Tukey post-hoc test with IBM SPSS 19.0 software (IBM, Armonk, NY, USA). Correlation analyses in software R version 3.5.2 (R Core Team, Vienna, Austria) using the package "corrplot" was conducted to determine the relationships between all variables. Pearson's correlations were conducted among the variables at the at the levels of $p < 0.01$ and 0.05.

Ordinary Kriging in Surfer 15 software (Golden Software Inc., Golden, CO, USA) was used to map the spatial distribution of soil water. The Semivariance ($\gamma(h)$) was adopted to identify the spatial variability of the vertical soil water distribution after the irrigation experiments using the ArcGIS 10.5 (ESRI, Redlands, CA, USA), which was expressed as follows:

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2 \tag{11}$$

where $N(h)$ was the sampled number of separated interval h, $Z(x_i)$ and $Z(x_i + h)$ were measured SWC at the location x_i and $x_i + h$, respectively. Three mathematical models (exponential, spherical and Gaussian models) were used for semivariance estimation. The optimal fit model was determined by the coefficient of determination (R^2) and the root-sum square (RSS) during the cross-validation. Four geostatistical parameters were generated based on the optimal fit model, including the effective range (A, cm), nugget (C_0), sill ($C_0 + C$) and the ratio of nugget to sill ($C_0/(C + C_0)$). The nugget (C_0) and partial sill (C) represented the variance caused by random error and the structural variance caused by systemic factors, respectively. The sill ($C_0 + C$), the sum of the C_0 and C, was the total variance of the SWC. A represented the distance at which the semivariance reached to the sill. The ratio of nugget to sill ($C_0/(C + C_0)$) indicated the spatial heterogeneity of the soil water distribution through the soil profile. Generally, the SWC was strongly spatially dependent if the ratio was less than 25%, moderately dependent if the ratio was between 25% and 75%, and weakly dependent if the ratio was higher than 75% (Jiang et al., 2015; Mei et al., 2018). Table 2 showed the description of the most variables used in this study.

3. Results

3.1. Soil properties, root traits and vegetated patch attributes

The differences in soil properties of VP and IP in the three states are shown in Table 3. A significant difference between VP and IP was observed for SOM within each site, although the differences were not significant for the other soil properties. Compared with IP, the SOM of VP in G, S1 and S2 were 9.24%, 16.74% and 9.09% greater, respectively. For both patches, the BD was significantly lower in S2 than in G and S1 ($p < 0.05$), and the ISWC in G was significantly lower than in S1 and S2 ($p < 0.05$). The SOM of IP in S2 was significantly higher than in the G and S1 ($p < 0.05$). The $WR_{0.25}$ of IP was significantly lower in S1 than in G and S2 ($p < 0.05$). But SOM and $WR_{0.25}$ did not differ among the VP in the three sites.

As shown in Fig. 3, plant roots at the three sites were mainly concentrated at a soil depth of 0 cm–20 cm and decreased along the soil profiles. Overall, the RLD_{fine} and RVD_{fine} of VP were greater than that of IP.

Table 2
Specification of the variables used in this study.

Variables	Description	Unit
MID	Maximum infiltration depth	cm
Unifr	Uniform infiltration depth	cm
DC	Dye coverage	%
PFP	Percentage of total infiltration flowing through paths of preferential flow	%
TDA	Total dyed area	cm^2
LI	Length index	
SWC_{diff}	Difference between initial soil water content and soil water content after irrigation	%
TIV	Total infiltration volume	mm
PIV	Preferential infiltration volume	mm
Con	Contribution of preferential infiltration volume to total infiltration volume	%
BD	Soil bulk density	$g\ cm^{-3}$
ISWC	Initial soil water content	%
SOM	Soil organic matter	$g\ kg^{-1}$
WR0.25	Soil water-stable aggregate content >0.25 mm	%
SWC	Soil water content	%
RLD	Root length density	$cm\ cm^{-3}$
RVD	Root volume density	$cm^3\ cm^{-3}$
RMD	Root mass density	$mg\ cm^{-3}$
Per	Vegetation patch perimeter	m
Com	Vegetation patch compactness	
AR	Geometric aspect ratio of vegetation patch	
AGB	Aboveground biomass	$g\ m^{-2}$
A	Range	cm
C_0	Nugget	
$C_0 + C$	Sill	
$C_0/(C + C_0)$	Relative structural variance	

The significant differences in RLD_{fine} and RVD_{fine} between VP and IP were found for the 0–20 cm soil depth in G and 0–10 cm in S2 ($p < 0.05$). Since G was dominated by *Bothriochloa ischaemum* characterized by a dense fibrous root system in upper soil layers, this site showed significantly higher RLD_{fine} at 0–20 cm soil depth and RVD_{fine} in 0–10 cm soil depth for both patches than RLD_{coarse} and RVD_{coarse} ($p < 0.05$). In comparison, significantly higher value of RVD_{coarse} at 0–10 cm depth were found in site S2 dominated by *Periploca* that has a taproot system. The RLD_{fine} in S1 for both patches were significantly greater than RLD_{coarse} through 0–40 cm depth ($p < 0.05$) and showed significantly greater than G and S2 in 30-40 cm ($p < 0.05$). Also, the RVD_{fine} in S1 at a depth > 20 cm were significantly greater than G and S2 ($p < 0.05$). The RMD was greater in VP through the soil profile than IP in three sites. A significant difference in RMD between VP and IP was only found at the 0-10 cm in S2 ($p < 0.05$). At 0-10 cm soil depth, the RMD of VP increased significantly from G to S2 following the natural vegetation succession ($p < 0.05$).

Plant communities within VP showed different attributes, such as plant height, size and biomass, during the different stages of natural vegetation succession (Table 4). Average patch height, *Com* and *AR* increased generally from G to S2, showing a positive relationship with duration of vegetation succession. VP length, width, area, *Per* and AGB in S1 were significantly greater than that in G and S2 ($p < 0.05$). Although S2 was characterized as having the most mature stage of vegetation succession, the patch area and *Per* of S2 at 0.14 m^2 and 1.14 m, respectively, were the lowest among the three sites investigated.

3.2. Extent of water infiltration

The extent of preferential flow was calculated using the dye stained images of the vertical soil profiles to interpreted subsurface water flow behavior (Fig. 4 and Table 5). A non-uniform distribution of stained area for all images was evident across a soil depth of 0 cm–40 cm. The dye-stained patterns under the same subplot exhibited some differences from any three replicates. The mean MID of VP was greater than that of IP in S1 and S2 by 35.73% and 23.43%, respectively, which indicated that water tended to infiltrate into deeper soil layers in VP than in IP at the two sites. But the mean MID of VP (24.63 cm) in G was slightly lower than that of IP (26.90 cm). The DC values of the soil profile for VP at all three sites were greater than those for IP. For both patches, the DC of S2 was lower than that of G and S1. Unifr represents the depth of matrix flow, and it was found to generally dominate the upper soil layer (0–10 cm). Most images showed that macropore flow interacted highly with preferential flow below a soil depth of 10 cm, appearing as lateral flow and finger flow (Weiler and Flühler, 2004; Allaire et al., 2009). The lowest mean values of Unifr were found for both patches in S2, with values of 11.87 cm for VP and 5.90 cm for IP respectively, followed by those at S1 with values of 12.30 cm and 11.50 cm, respectively. However, no significant differences were detected in these parameters for the two patches and different vegetation sites ($p > 0.05$).

3.3. Spatial variation of soil water content in vertical soil profile

The spatial distribution of difference between ISWC and soil water content after irrigation (SWC_{diff}) through the soil profile was revealed by the contour maps (Fig. 5). Areas of high and low water content indicating paths of macropore flow with interaction with the surrounding soil matrix were distributed randomly in the contour maps. In general, higher SWC_{diff} appearing as a belt were more frequently evident in the maps of IP. By contrast, regions with both higher and lower SWC appeared more frequently with a scattered distribution in VP. Semivariogram analysis further indicated the degree of spatial variability of the distribution of SWC_{diff} (Table 6). The relative structural variance $[C_0/(C + C_0)]$ of SWC_{diff} was <25%, indicating that SWC in all soil sections had obvious spatial dependence. The mean spatial autocorrelation ranges (A) of VP for the three sites were greater than that of IP, indicating that SWC was more variable in VP. However, the differences in the semivariogram parameters between VP and IP and among three sites were not relevant.

Fig. 6 shows the water infiltration volume after irrigation across the vertical soil section. VP in all three sites appeared to have more infiltration volume than IP, in particular below the depth of the Unifr where

Table 3
A comparison of soil properties (0–40 cm soil depth) of vegetation patches (VP) and interspace patches (IP) in the three study sites.

Site	Patch types	BD (g/cm^3)	ISWC (%)	SOM ($g\ kg^{-1}$)	$WR_{0.25}$ (%)	Particle size composition (%)		
						Clay (<0.002 mm)	Silt (0.002–0.02 mm)	Sand (0.02–2 mm)
G	VP	1.23 ± 0.01aA	9.45 ± 0.68aB	5.32 ± 0.32aA	47.35 ± 2.09aA	11.80 ± 0.30aA	25.47 ± 0.24aA	63.73 ± 0.27aA
	IP	1.22 ± 0.01aA	9.34 ± 0.06aB	4.87 ± 0.05bB	46.64 ± 1.62aA	11.25 ± 0.01aA	24.62 ± 0.54aA	65.13 ± 0.51aA
S1	VP	1.25 ± 0.01aA	12.27 ± 0.12aA	5.37 ± 0.30aA	43.12 ± 6.09aA	10.75 ± 0.07aA	23.49 ± 0.16aA	65.76 ± 0.98aA
	IP	1.27 ± 0.01aA	11.02 ± 0.08aA	4.60 ± 0.05bB	39.44 ± 1.93aB	10.80 ± 0.01aA	23.55 ± 0.09aA	65.66 ± 0.16aA
S2	VP	1.12 ± 0.03aB	10.42 ± 0.09aA	5.76 ± 0.16aA	47.48 ± 1.19aA	11.77 ± 0.14aA	23.51 ± 0.74aA	64.72 ± 0.84aA
	IP	1.14 ± 0.02aB	10.68 ± 0.01aA	5.28 ± 0.08bA	48.26 ± 5.43aA	11.54 ± 0.12aA	25.23 ± 1.09aA	64.23 ± 1.58aA

Note: Data represent means ± standard deviation (n = 3 replications × 4 soil layers = 12). Different lowercase letters in one column represent significant difference ($p < 0.05$) between patches within a site, and different uppercase letters in one column mean significant difference ($p < 0.05$) among different three sites for the same patch type.

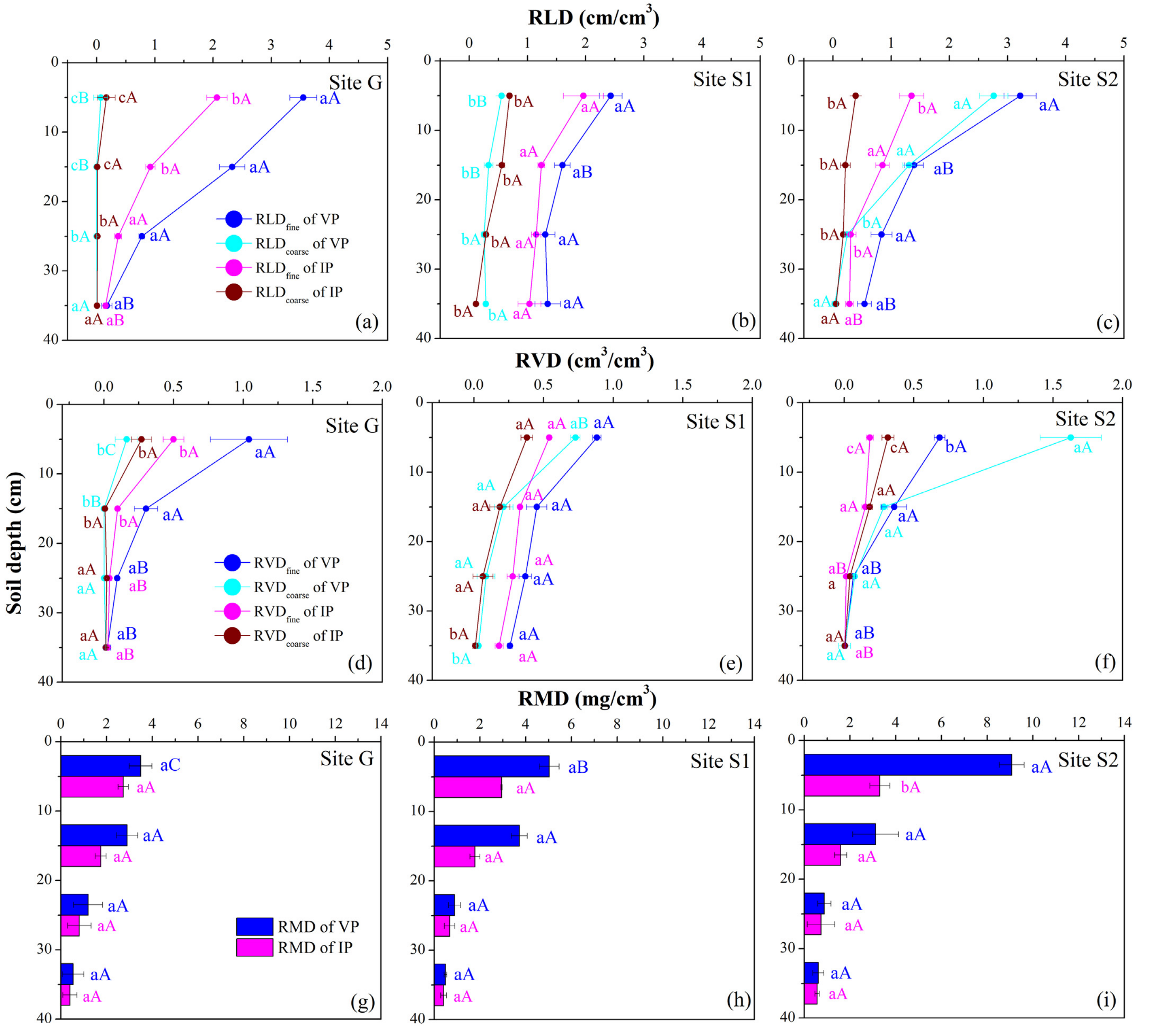

Fig. 3. Root length density of fine roots (RLD_{fine}) and coarse roots (RLD_{coarse}) in soil profiles for vegetation patches and interspace patches at G (a), S1 (b) and S2 (c); Root volume density of fine roots (RVD_{fine}) and coarse roots (RVD_{coarse}) in soil profiles at G (d), S1 (e) and S2 (f); Root mass density (RMD) in soil profiles at G (g), S1 (h) and S2 (i). Different lowercase letters represent significant difference ($p < 0.05$) among different roots of patches at the same soil layer within a site, and different uppercase letters mean significant difference ($p < 0.05$) among different three sites for the same patch type at the same soil layer.

preferential flow occurred. There was a significant difference between patches at the depth of 22-24 cm and 26-28 cm in G and the depth of 20-24 cm in S1($p < 0.05$). The infiltration in some soil layers below Unifr was greater than that above, indicating that more water could transport into the deep soil layers through preferential pathways. Among the three sites, VP of G had significantly lower infiltration at topsoil (0-4 cm soil depth) than that of S1 and S2. There were no significant differences in PIV and Con between VP and IP among the three sites (Fig. 7), but high values of PIV were observed for VP. Expect for S2, the Con was larger for VP relative to that for IP in G and S1. For VP, the

Table 4
Comparison of vegetation patch attributes among the three sites.

Site	Patch type	Height (m)	Length (m)	Width (m)	Area (m^2)	Per (m)	Com	AR	AGB ($g\ m^{-2}$)
G	*Bothriochloa ischaemum*	0.58 ± 0.03 b	0.69 ± 0.09 b	0.63 ± 0.09 b	0.45 ± 0.104 b	2.09 ± 0.21 a	0.89 ± 0.01 a	1.11 ± 0.06 a	178.43 ± 16.85 b
S1	*Artemisia sacrorum*	0.73 ± 0.11 a	0.98 ± 0.11 a	0.81 ± 0.16 a	0.81 ± 0.126 a	2.81 ± 0.47 a	0.89 ± 0.01 a	1.23 ± 0.15 a	269.79 ± 33.64 a
S2	*Periploca sepium Bunge*	0.78 ± 0.06 a	0.45 ± 0.12 c	0.28 ± 0.18 c	0.14 ± 0.085 c	1.14 ± 0.29 a	0.93 ± 0.05 a	1.751 ± 0.07 a	232.48 ± 15.02 b

Note: Data represent means ± standard deviation (n = sample numbers in plots). Different lowercase letters in one column represent significant difference at $p < 0.05$ among three sites. Per represents the patch perimeter, Com represents the vegetation patch compactness; AR represents the geometric aspect ratio of vegetation patch, AGB represents the aboveground biomass.

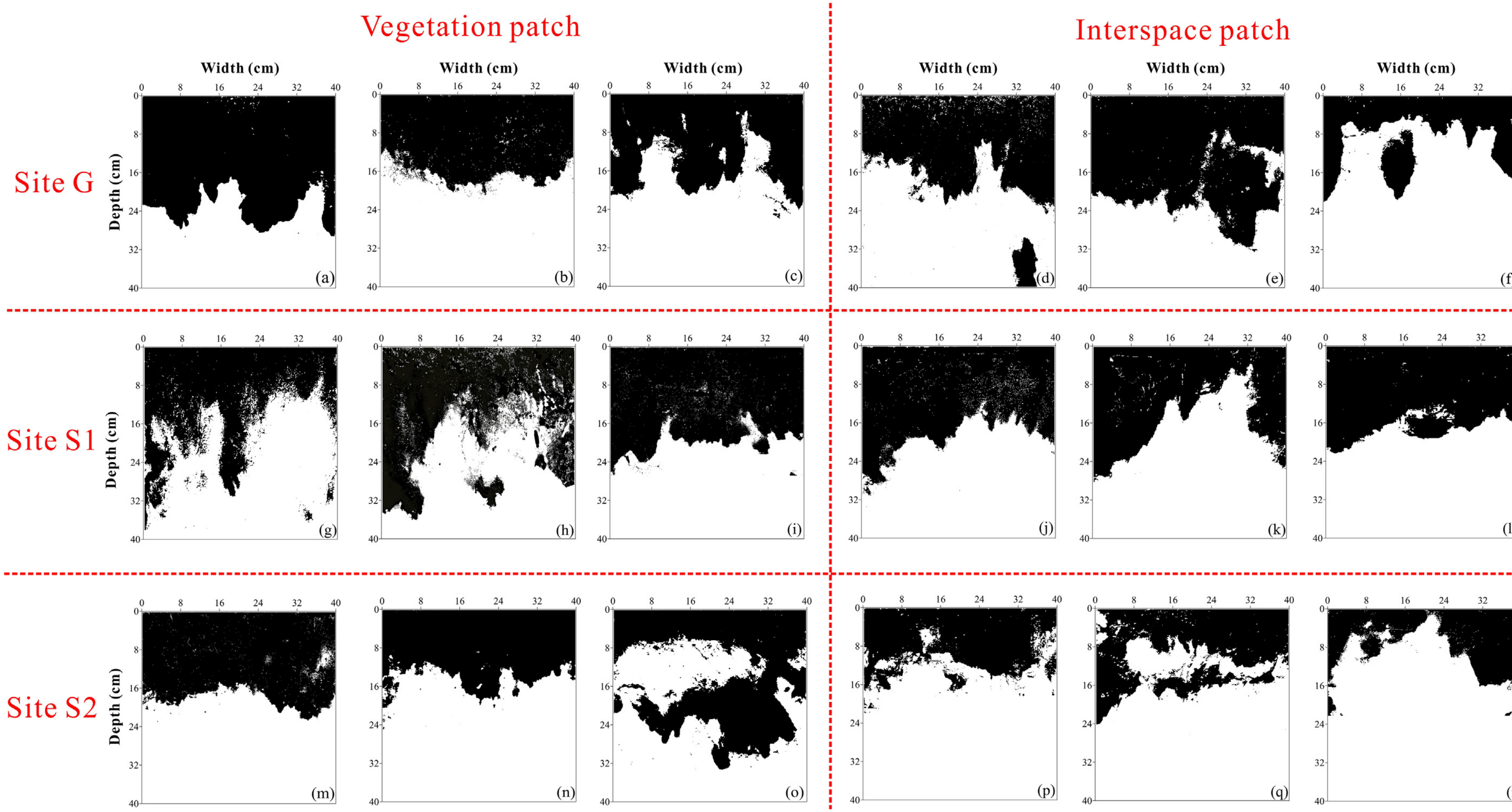

Fig. 4. Infiltration patterns as revealed by dye stained images of the vertical soil profiles. (a)-(c) and (d)-(f) show the infiltration patterns for three subplots under vegetation patches (VP) and interspace patches (IP) at site G, respectively. (g)-(i), (j)-(i), (m)-(o), and (p)-(r) are same as (a)-(c) and (d)-(f), but for site S1 and S2, respectively.

Table 5
Water flow parameters as determined by dye stained images of the vertical soil profile.

Site	Patch	Subplot	MID	Unifr	DC	FFP	LI
			(cm)	(cm)	(%)	(%)	
G	VP	1	29.20	18.80	58.31	19.40	398.00
		2	20.80	15.90	40.72	2.38	334.00
		3	23.90	9.10	39.39	42.24	347.00
		Mean	24.63 ± 2.45aA	14.60 ± 2.87aA	46.14 ± 6.10aA	21.34 ± 11.55aA	359.67 ± 19.63aA
	IP	1	25.90	13.00	44.26	26.57	402.00
		2	32.80	21.20	55.44	4.40	361.50
		3	22.00	4.50	23.88	52.89	185.25
		Mean	26.90 ± 3.16aA	12.90 ± 4.82aA	41.19 ± 9.24aA	27.95 ± 14.01aA	316.25 ± 66.53aA
S1	VP	1	37.90	9.80	42.23	41.98	313.00
		2	35.90	11.90	50.81	41.44	244.00
		3	26.50	15.20	44.99	15.54	383.00
		Mean	33.43 ± 3.51aA	12.30 ± 1.57aA	46.01 ± 2.53aA	32.99 ± 8.72aA	313.33 ± 40.12aA
	IP	1	29.50	15.00	43.86	14.50	436.75
		2	24.00	8.10	37.21	45.58	242.00
		3	20.40	11.40	37.15	23.28	226.75
		Mean	24.63 ± 2.65aA	11.50 ± 1.99aA	39.41 ± 2.23aA	27.79 ± 9.25aA	301.83 ± 67.60aA
S2	VP	1	22.80	16.50	44.60	7.51	399.00
		2	24.70	12.40	36.26	14.51	394.00
		3	33.10	6.70	44.19	62.10	278.00
		Mean	26.87 ± 3.16aA	11.87 ± 2.84aA	41.68 ± 2.71aA	28.04 ± 17.15aA	357.00 ± 39.52aA
	IP	1	18.00	9.00	28.12	19.99	386.00
		2	25.30	5.90	35.51	41.54	331.25
		3	22.00	2.80	22.65	30.91	291.75
		Mean	21.77 ± 2.11aA	5.90 ± 1.79aA	28.76 ± 3.73bA	30.81 ± 6.22aA	336.33 ± 27.33aA

Note: Data represent means ± standard deviation (n = 3 replications). MID means the maximum infiltration depth, Unifr means the uniform infiltration depth, DC means the dye coverage to the total soil profile region, FFP means the preferential flow proportion, LI means the length index. Different lowercase letters in one column denote significant difference ($p < 0.05$) between patches within a site, and different uppercase letters in one column denote significant difference ($p < 0.05$) among different three sites for the same patch type.

mean PIV (28.53 mm) and Con (60.58%) in S1 were significantly higher than in G and S2 ($p < 0.05$). The IP in G had significantly lower PIV (15.48 mm) and Con (37.81%) than in S1 and S2 ($p < 0.05$).

3.4. Factors driving preferential flow

Correlation analyses were used to investigate the relationship between water flow parameters and their driving factors (Fig. 8). From the perspective of soil and root properties, RLD_{fine} and RVD_{fine} appeared to have the most influence compared to the other factors. RLD_{fine} and RVD_{fine} had significantly positive relationships with MID, DC, FFP, PIV and SWC_{diff} ($p < 0.01$), whereas they had negative relationships with Unifr ($p < 0.05$). For vegetation patch attributes, patch area showed the strongest positive relationships with the most water flow parameters (MID, PFP, SWC_{diff}, PIV, and Con) ($p < 0.01$).

4. Discussion

4.1. Comparison of vegetation patches and interspace patches

The present study demonstrated that VP improved the underlying root system (Fig. 3) and influenced soil properties (Table 3). The dye tracing experiment also demonstrated that VP changed infiltration behavior. The calculated preferential flow indices including MID, Unifr, DC, FFP and LI from the dye profile combined with measured SWC_{diff}, PIV and Con can be used to quantify the extent of soil water flow and the degree of preferential flow (Jiang et al., 2015; Wang et al., 2019). Although diverse vertical soil water flow behaviors were evident among experimental plots (Fig. 4), the degree of preferential flow in VP was generally greater than that in IP. This result is consistent with findings of many previous studies (Peng et al., 2013; Eldridge et al., 2015; Meron, 2016; Vandandorj et al., 2017), which reported significantly greater water infiltration capacities for shrub patches than for interspace or open areas. Preferential flow behaviors are generally affected by terrain attributes, soil structure, permeability to soil water, faunal activities, root system features, moisture condition and management practices (Benegas et al., 2014; Schneider et al., 2018). At the patch scale, VP and IP exhibited different SOM, RLD, RVD and RMD (Table 3 and Fig. 3). Hu et al. (2019) using an X-ray computed method found that the high number of soil macropores under shrubland was closely associated with high root density. In the present study, RLD_{fine} and RVD_{fine} were found to have greater effects on increasing preferential flow compared to root mass, coarse root length and soil properties (Fig. 8), which is consistent with observations from previous studies (Cui et al., 2019; Luo et al., 2019). Zhang et al. (2015) similarly found that RLD and RMD were greater in preferential pathways than in the soil matrix. For both VP and IP, fine roots accounted for >80% of total roots across the 0 cm–40 cm soil section (Fig. 3). The larger quantity of fine roots provided a greater interface for root-soil contact, and the roots form an interconnected network and non-capillary channels that enhance water infiltration (Sidle et al., 2001). In addition, the rhizosphere exudates of fine roots can increase SOM, subsequently contributing to soil aggregation and the improvement of soil physical properties (Bronick and Lal, 2005).

Although VP exhibited greater degree of preferential flow than IP, there were no significant differences in water flow parameters between the two patches (Table 5). In the Inner Mongolia grassland of northern China, shrub patches had significantly greater water infiltration rates and deeper maximum wetting depths than the interspace patches (Peng et al., 2013). Hu et al. (2019) further investigated soil macroporosity and found that it was significant higher in soil under the interspace grass than under the shrub patch at most sites. Benegas et al. (2014) found that tree density affects the degree of preferential flow, with smaller interspace areas demonstrating greater preferential flow than larger interspace areas. The densities of VP in the sloping grassland of the Loess Plateau (Table 1) were greater than those in the Inner Mongolia grassland (Peng et al., 2013; Hu et al., 2019), indicating that the IP in our loess slope with smaller size had relatively great preferential flow. Moreover, the effect of patchy vegetation on soil macroporosity may extend beyond the canopy edge of individual plants due to the influence of the plant roots (Eldridge et al., 2015). The present study identified that the root volume and root length in the soil profile of IP for all three sites were high, despite they were lower than those of VP. During vegetation dynamics and ecosystem processes, the plants

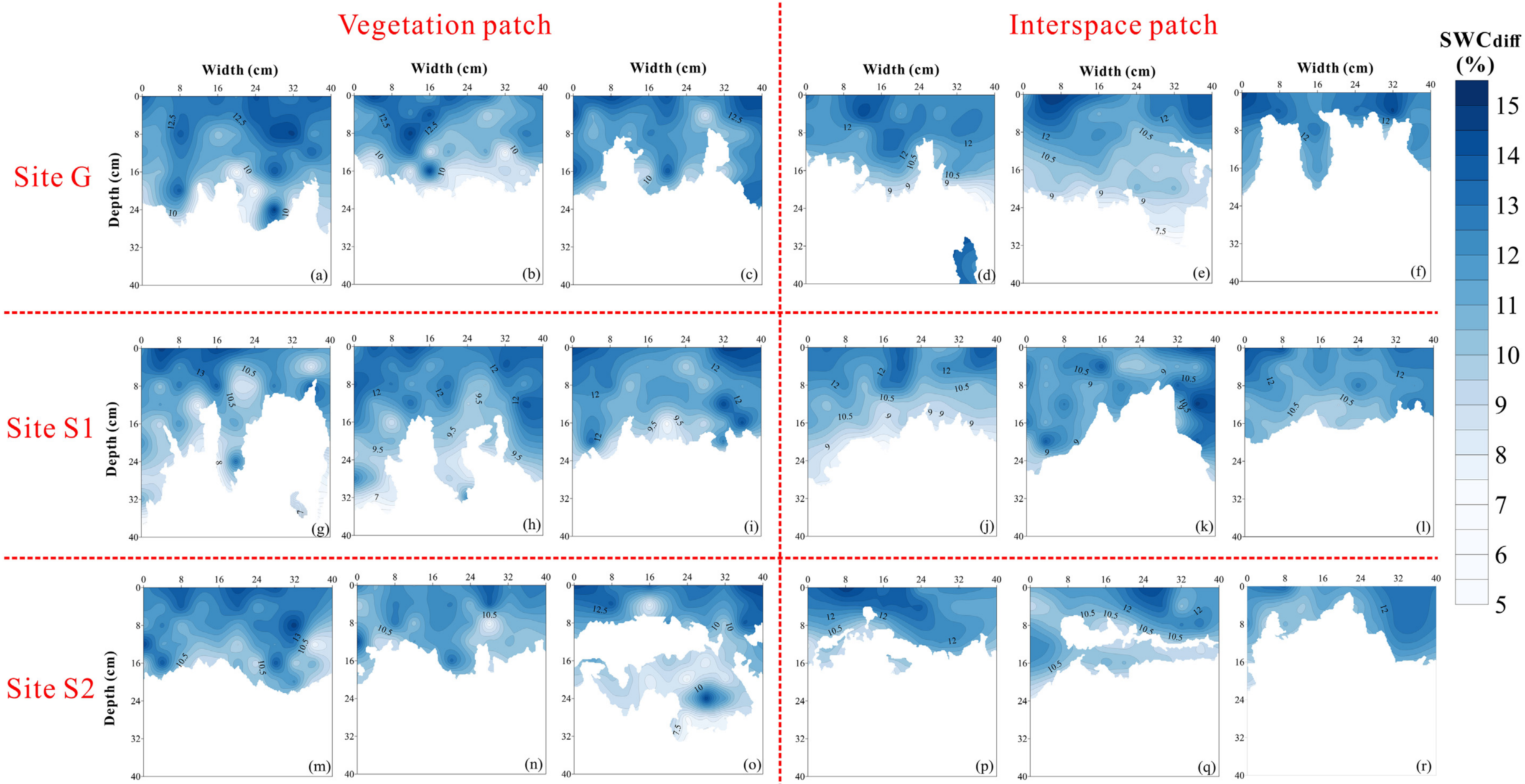

Fig. 5. Spatial distribution of soil water content (%) in the vertical soil sections for two patches in the three sites. (a), (b) and (c) correspond to the vegetation patches at site G, respectively. (d), (e) and (f) correspond to the interspace patches at site G, respectively. (g), (h) and (i) correspond to the vegetation patches at site S1, respectively. (j), (k) and (i) correspond to the interspace patches at site S1, respectively. (m), (n) and (o) correspond to the vegetation patches at site S2, respectively. (p), (q) and (r) correspond to the interspace patches at site S2, respectively.

Table 6
Semivariogram parameters showing the degree of spatial variability of the distribution of soil water content.

Site	Patch	Test	Model	SWC_{diff} (%)	A (cm)	Nugget (C_0)	Sill (C_0 + C)	$C_0/(C + C_0)$ (%)	R^2	RSS
G	VP	1	Spherical	10.10	48.15	0.12	4.49	2.61	0.99	0.08
		2	Exponential	9.02	41.48	0.11	7.39	1.47	0.99	0.04
		3	Exponential	9.78	32.98	0.07	3.04	2.19	0.99	0.04
		Mean		9.63 ± 0.32aA	40.87 ± 4.38aA	0.10 ± 0.07aA	4.98 ± 1.28aA	2.09 ± 0.33aA		
	IP	1	Spherical	9.34	49.57	0.06	3.65	1.60	0.91	0.01
		2	Spherical	9.35	46.37	0.05	4.76	0.97	0.92	0.01
		3	Gaussian	9.29	26.62	0.06	1.24	5.23	0.85	0.02
		Mean		9.33 ± 0.02aA	40.85 ± 7.98aA	0.06 ± 0.01aA	3.22 ± 1.04aA	2.60 ± 1.32aA		
S1	VP	1	Exponential	9.72	56.57	0.0643	4.80	1.34	0.99	0.05
		2	Exponential	9.78	43.12	0.1902	6.01	3.16	0.99	0.04
		3	Spherical	9.55	40.69	0.0709	5.53	1.28	0.99	0.04
		Mean		9.68 ± 0.07aA	46.79 ± 4.93aA	0.11 ± 0.04aA	5.45 ± 0.35aA	1.93 ± 0.62bA		
	IP	1	Spherical	9.06	42.21	0.1545	3.64	4.25	0.93	0.01
		2	Spherical	9.47	19.81	0.1577	1.52	10.36	0.86	0.02
		3	Spherical	9.19	54.57	0.0651	2.27	2.87	0.89	0.01
		Mean		9.24 ± 0.12aA	38.86 ± 9.70aA	0.13 ± 0.03aA	2.48 ± 1.07aA	5.82 ± 2.30aA		
S2	VP	1	Spherical	9.95	49.44	0.1154	8.97	1.29	0.99	0.04
		2	Exponential	9.76	51.48	0.149	7.39	2.02	0.99	0.04
		3	Spherical	9.22	38.98	0.01366	3.04	0.45	0.99	0.04
		Mean		9.64 ± 0.22aA	46.63 ± 3.87aA	0.09 ± 0.04aA	6.47 ± 1.77aA	1.25 ± 0.45aA		
	IP	1	Spherical	9.29	56.57	0.0064	5.42	0.12	0.90	0.01
		2	Gaussian	9.12	36.93	0.0074	1.93	0.38	0.91	0.01
		3	Spherical	8.38	32.13	0.126	3.13	4.03	0.86	0.01
		Mean		8.93 ± 0.28bA	41.88 ± 11.65aA	0.05 ± 0.04aA	3.49 ± 1.02aA	1.51 ± 1.26aB		

Note: RSS means the residual sum of squares. The mean value of each site represents mean ± SD, $n = 3$. The mean value of each site represents mean ± SD, n = 3. The different lowercase letters for mean value in one column represent significant difference at $p < 0.05$ between patches within a site, and different uppercase letters in one column denote significant difference at $p < 0.05$ among different three sites for the same patch type.

migrate along the hillslope and may colonize in IP and retreat in VP (Saco et al., 2007; Deblauwe et al., 2012; Wu et al., 2016). Even though the previous plants have been dead within IP for a long time, their decayed or decaying roots persist in forming continuous preferential flow pathways to facilitate water movement (Guo et al., 2019).

The well-developed preferential flow paths lead to high spatial variation in SWC of the soil profile (Shinohara and Otsuki, 2015). This study showed that no significant differences were found in spatial heterogeneity of SWC between VP and IP, but VP tended to have more variability (Fig. 5 and Table 6). SWC measured in all soil layers after irrigation for both patches increased compared to ISWC ($SWC_{diff} > 0$), and mean SWC_{diff} decreased with increasing soil depth generally (Fig. 5). This indicated that gravity was the driver of downward water movement in these sites. Nevertheless, the infiltration volume increased with increasing soil depth for certain soil layers (Fig. 6) along with the random scatter of high and low SWC in the Kriging maps (Fig. 5). It

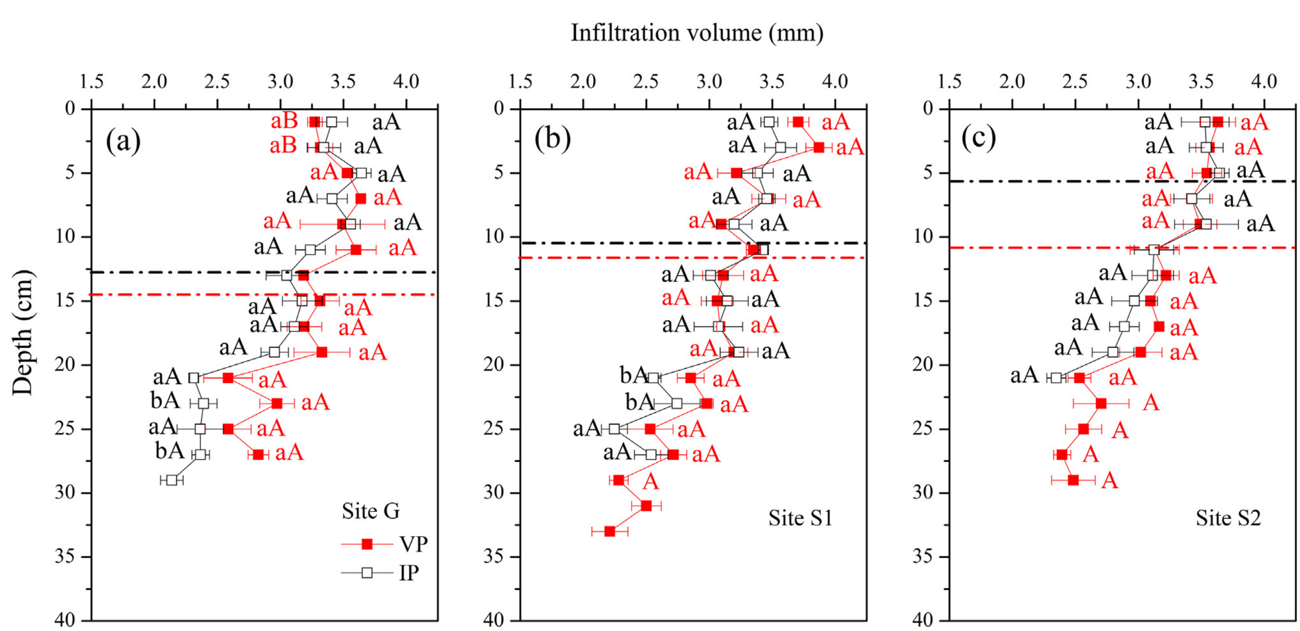

Fig. 6. Soil water infiltration volume with soil depth through the vertical profile for VP and IP in site G (a), S1 (b) and S2 (c). The red horizontal dotted line in figure represents the average Unifr for VP, and the black horizontal dotted line represents the average Unifr for IP. Different lowercase letters represent significant difference ($p < 0.05$) between VP and IP at the same soil depth within a site, and different uppercase letters indicate significant difference ($p < 0.05$) among different three sites for the same patch type at the same soil depth. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

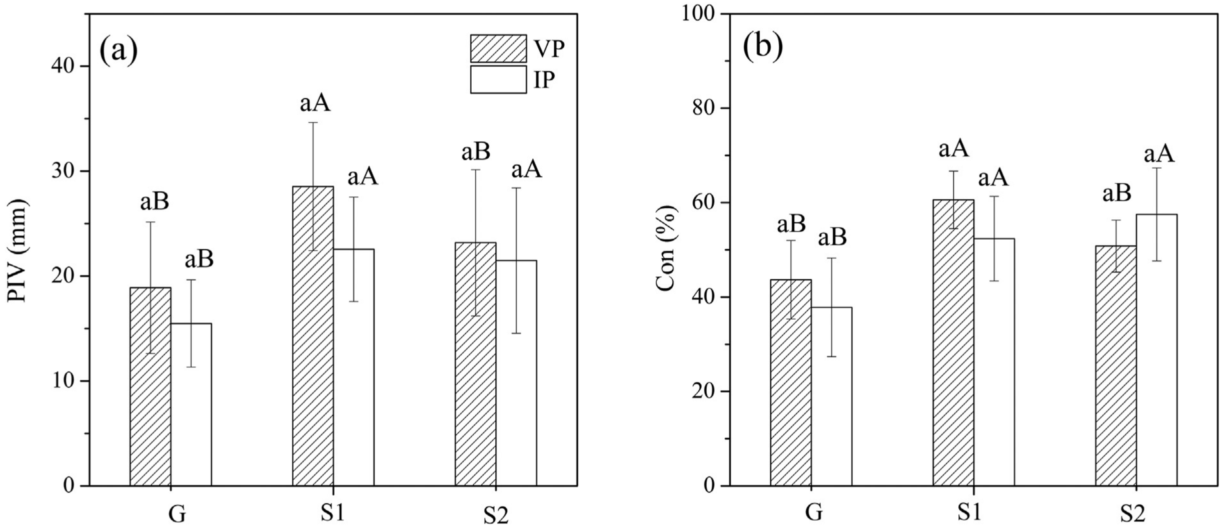

Fig. 7. Average preferential infiltration volume (PIV, mean ± SE, $n = 3$) (a) and their contribution to the total infiltration volume (Con, mean ± SE, $n = 3$) (b) for the vegetation patches and interspace patches in the three sites. Different lowercase letters above the bars represent significant difference ($p < 0.05$) between VP and IP within a site, and different uppercase letters above the bars indicate significant difference ($p < 0.05$) among the three sites for VP or IP.

was revealed that gravity was not the only driver of water infiltration. Water exchange between preferential flow pathways and the surrounding soil matrix is significant when water flows downward (Cey and Rudolph, 2009; Laine et al., 2014). Based on the representation of dye concentration for water exchange (Jiang et al., 2015), the high and low values of the SWC distribution isograms were considered to represent strong and weak water interactions between the two zones, respectively. Beven and Clarke (1986) noted that water transfer from preferential pathways to the soil matrix is also referred to as "lateral infiltration from the macropores", which is crucial during water exchange. Previous studies reported that root systems with relatively large and shallow roots that are well connected with plant stems can facilitate lateral flow (Luo et al., 2019; Guo et al., 2020). Thus, the more abundant and intricate roots with a lateral direction especially under VP likely contributed to the lateral flow and spatial variability of SWC.

Vegetation patchiness changed not only the spatial patterns of infiltration but also the infiltration amount. Fig. 6 shows that the infiltration amount did not always decrease with increasing soil depth, particularly for certain soil layers below Unifr in which preferential flow occurred. PIV can be effectively used to separate the contribution of preferential flow from water infiltration (Wang et al., 2019). Previous studies have shown that preferential flow paths and large macrospores can carry the majority of water after soil becomes saturated (Bogner et al., 2010; Singh et al., 2015). In the present study, VP with a higher degree of preferential flow had higher PIV compared to IP (Fig. 7). This result is in agreement with the results of Mei et al. (2018), who found that plantations and natural grassland in the upper slopes of the Loess Plateau had lower PIV than natural forestland in which there was higher preferential flow.

4.2. Comparison among the states of vegetation succession

Different plant species displaying particular root morphometric features can induce different water infiltration behaviors (Yu et al., 2016;

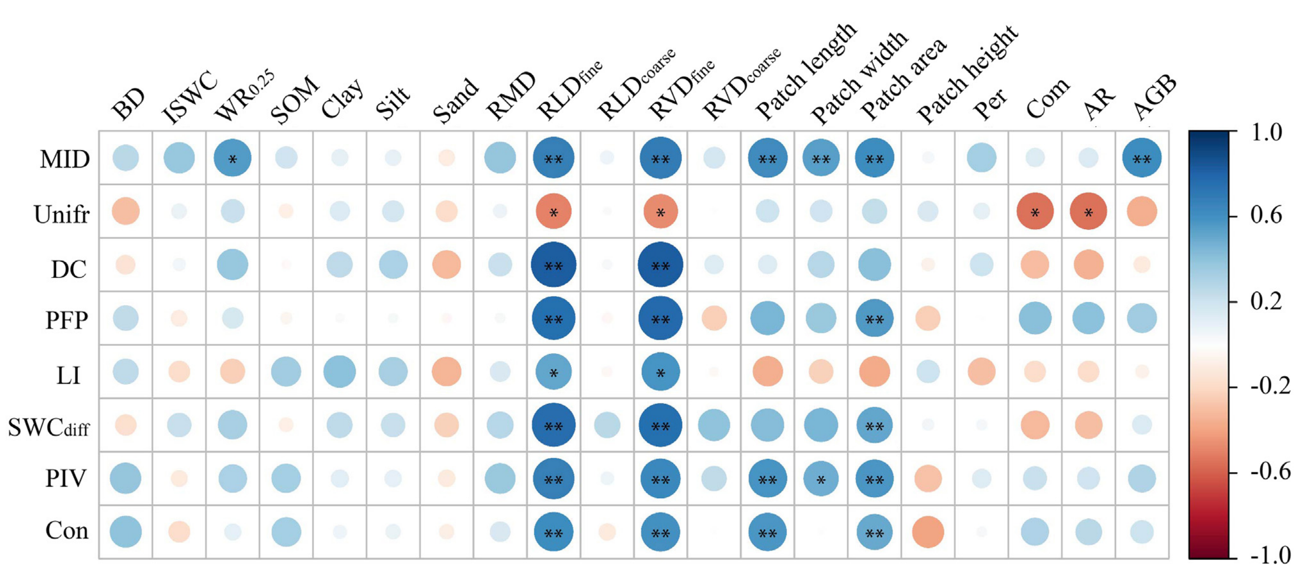

Fig. 8. Correlation analysis of water flow parameters (MID, Unifr, DC, PFP, LI, SWC_{diff}, PIV and Con) and impacting factors that include soil properties (BD, ISWC, $WR_{0.25}$, SOM, Clay, Silt and Sand), root characteristics (RMD, RLD_{fine}, RLD_{coarse}, RVD_{fine} and RVD_{coarse}) and vegetation patch attributes (length, width, area, height, Per, Com, AR and AGB). * represents significant correlation at the 0.05 level ($P < 0.05$), and ** represents significant correlation at the 0.01 level ($P < 0.01$).

Jiang et al., 2018; Zhang et al., 2019a). Contrary to our hypothesis, the degree of preferential flow did not increase with increasing vegetation succession. As shown in Fig. 7, the VP of S1 exhibited significantly greater PIV and Con than that of G and S2. As previously mentioned, fine roots in the soil profile played a vital role in water infiltration behaviors (Fig. 8). The significantly highest values of RLD_{fine} and RVD_{fine} were found below Unifr where preferential flow pathways were identified in S1 (Fig. 3). Previous studies indicated that the contents of plant roots were high in preferential pathways, thereby facilitating water flow (Bogner et al., 2010; Zhang et al., 2015). Additionally, the dominant vegetation type in S2 of *P, sepium Bunge* had higher RLD_{coarse} and RVD_{coarse} than *B. ischaemum* in G and *A. sacrorum* in S1. Cui et al. (2019) suggested that non-perishable coarse roots that compact soil and clog soil pore space could block water flow and inhibit soil infiltration process.

Table 3 shows that S1 exhibited greater vegetation patch length, width, area, *Per* and AGB compared to G and S2. Hao et al. (2016) found that soil water content increases with increasing shrub patch size on the Loess Plateau. Similarly, our study indicated that patch area were significantly and positively related to the degree of preferential flow (Fig. 8). It is expected that a larger shrub vegetation patch would have deeper roots and would generate a greater mass of litter, which could promote infiltration through increasing aggregation and soil organic matter (Zhang et al., 2018; Marquart et al., 2019). The large shrub canopy provides habitat and nutrition for termites, ants and earthworms, and these organisms act as ecosystem engineers because they construct a great number of biopores and tunnels in the soil surface, subsequently increasing infiltration rates (Sarr et al., 2001). On the other hand, the vegetation patch attributes determine the microtopography within a patch and play a vital role in water retention and infiltration (Rossi and Ares, 2017). *A. sacrorum* present in S1 showed the largest patch size with multiple and diffused stems at the base, which increased surface roughness and could result in more variable water infiltration behaviors. In contrast, *P. sepium Bunge* present in S2 characterized by a main stem and became established for just a few years, and *B. ischaemum* at G had dense and concentrated tillers at the base.

4.3. Implications and limitations of the present study

The present study found that IP had lower infiltration capacity and preferential flow, and therefore is a source of surface runoff and sediment, whereas VP has greater infiltration capacity and preferential flow, and therefore serves as a sink. The greater infiltrability and preferential flow of VP has been found to be beneficial for retaining surface runoff and sediment in semiarid grassland (Wang et al., 2019). Meron (2016, 2018) indicated that the infiltration contrast between VP and IP induced overland water flow towards areas of VP, which enhanced the water transport for VP and further accelerated the vegetation growth (Getzin et al., 2016). On the other hand, VP increases soil heterogeneity and was therefore not expected to be associated with general reductions in ecosystem functions and processes (Eldridge et al., 2011; Muñoz-Robles et al., 2011). However, differences in preferential flow between VP and IP were found to be not significant. Considering the five stages of shrub encroachment (D'Odorico et al., 2012), the mosaic landscape in the present study was considered as "grassland with shrubs". At this stage, increases in shrub density did not occur at the expense of grass cover, with small heterogeneity between them (Okin et al., 2009). This confirmed that IP was able to compete with VP for infiltration, accumulating nutrients and retaining surface runoff and sediment. Eldridge et al. (2015) indicated that retention of vegetation cover and less disturbance in IP was crucial for maximizing the capture of water and increasing infiltration on the whole hillslope scale. This mosaic landscape with small heterogeneity between VP and IP in our study might contribute to increasing general infiltration, reducing erosion and maintaining ecosystem stability along the Loess slopes. Furthermore, the greatest preferential flow occurred in S1 among the three states of vegetation succession. Therefore, abandoned farmland undergoing natural restoration by native plants, especially subshrub, might be regarded as an applicable approach for ecological conservation of the Loess Plateau.

Although dye trace experimentation is a common approach used to study water flow patterns, this method is mainly limited to two-dimensional (2D) detection. Future research should preferably examine the detailed images of macropore characteristics through X-ray computed tomography (CT). In addition, the present study placed an emphasis on comparing the effects of VP and IP on subsurface water flow. However, upscaling the results of this patch-scale study to estimate infiltration patterns at the hillslope or watershed scale may not be appropriate.

5. Conclusions

The present study has evaluated the effect of vegetation patchiness on vertical soil water behavior during natural restoration in semiarid grass slopes. The results showed that root length density and root volume density of fine roots (RLD_{fine} and RVD_{fine}) were significantly correlated with preferential flow, and are therefore the dominant drivers of water flow. VP tended to exhibit greater preferential flow and preferential infiltration than IP. However, these differences between the two patches were not significant due to the presence of considerable numbers of decayed roots and fine roots under IP. The three vegetation succession states for VP showed the greatest PIV (28.53 mm) and Con (60.58%) in the S1. Among all vegetation patch attributes, patch size had a significantly positive relationship with preferential flow. The results of the present study demonstrated that a mosaic landscape in a native grass slope has small infiltration heterogeneity that is beneficial for the stabilization of ecosystem functions. Subshrub vegetation can be considered as an optimum state of vegetation restoration for abandoned farmland of the Loess Plateau.

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CRediT authorship contribution statement

Rui Wang: Conceptualization, Methodology, Writing - original draft. **Zhibao Dong:** Conceptualization, Supervision. **Zhengchao Zhou:** Writing - review & editing. **Ning Wang:** Software, Investigation. **Zhijing Xue:** Visualization. **Liguo Cao:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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